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Marine power supply system and method for controlling marine power supply system

Abstract

A marine power supply system includes a first battery to supply power to an electric motor, a battery switch to be manually switched between an on-state in which the first battery is used and an off-state in which the first battery is not used, and a notifier to provide a notification indicating that the battery switch is in the off-state when the battery switch is in the off-state.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application No. 2020-058945 filed on Mar. 27, 2020. The entire contents of this application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a marine power supply system and a method for controlling a marine power supply system.

2. Description of the Related Art

(3) A marine power supply system that supplies power to a marine propulsion unit including an engine and an electric motor and a method for controlling a marine power supply system are known in general. Such a marine power supply system and a method for controlling a marine power supply system are disclosed in Japanese Patent Laid-Open No. 2017-218016, for example.

(4) Japanese Patent Laid-Open No. 2017-218016 discloses a marine vessel including a marine propulsion device including an engine and an electric motor. The marine propulsion device includes a propeller. Both the engine and the electric motor generate power to rotate the propeller. The marine vessel includes a battery to supply power to the electric motor. The electric motor rotates the propeller using the power from the battery.

(5) Although not described in Japanese Patent Laid-Open No. 2017-218016, the marine vessel includes a battery switch to be manually switched between a state in which the battery is used (hereinafter referred to as an “on-state”) and a state in which the battery is not used (hereinafter referred to as an “off-state”) in order to significantly reduce or prevent discharge from the battery due to dark current, for example, during storage of the marine vessel. In a marine vessel including an electric motor as disclosed in Japanese Patent Laid-Open No. 2017-218016, a battery switch is conceivably provided between the electric motor and a battery. However, conceivably in the marine vessel as described above, when the engine is driven, the propeller rotates to propel the marine vessel only by driving the engine. Thus, the marine vessel conceivably travels without a user being aware that the battery switch is in an off-state. Therefore, a marine power supply system and a method for controlling a marine power supply system that significantly reduce or prevent traveling of a marine vessel without a user being aware that a battery switch is in an off-state even when power is supplied to a marine propulsion device (marine propulsion unit) including an engine and an electric motor have been desired.

SUMMARY OF THE INVENTION

(6) Preferred embodiments of the present invention provide marine power supply systems and methods for controlling marine power supply systems that each significantly reduce or prevent traveling of marine vessels without users being aware that battery switches are in off-states even when power is supplied to marine propulsion units including engines and electric motors.

(7) A marine power supply system according to a preferred embodiment of the present invention supplies power to a marine propulsion unit including a propulsion generator that generates a propulsive force to propel a marine vessel, an engine that drives the propulsion generator, and an electric motor that drives the propulsion generator, and includes a first battery to supply power to the electric motor, a battery switch to be manually switched between an on-state in which the first battery is used and an off-state in which the first battery is not used, and a notifier to provide a notification indicating that the battery switch is in the off-state when the battery switch is in the off-state.

(8) A marine power supply system according to a preferred embodiment of the present invention includes the notifier to provide the notification indicating that the battery switch is in the off-state when the battery switch is in the off-state. Accordingly, the notifier notifies a user that the battery switch is in the off-state. Consequently, even when power is supplied to the marine propulsion unit including the engine and the electric motor, traveling of the marine vessel without the user being aware that the battery switch is in the off-state is significantly reduced or prevented.

(9) A marine power supply system according to a preferred embodiment of the present invention preferably further includes a controller configured or programmed to control the notifier to provide the notification indicating that the battery switch is in the off-state when the battery switch is in the off-state. Accordingly, the notifier is easily controlled using the controller.

(10) In such a case, the notifier preferably includes a display, and the controller is preferably configured or programmed to control the display to provide a display indicating that the battery switch is in the off-state when the battery switch is in the off-state. Accordingly, the user easily

recognizes that the battery switch is in the off-state by visually recognizing the display.

(11) In a marine power supply system including the display, the controller is preferably configured or programmed to control the display to display message information indicating that the battery switch is in the off-state when the battery switch is in the off-state. Accordingly, the user easily recognizes that the battery switch is in the off-state by reading the message information on the display.

(12) In a marine power supply system including the controller, the controller is preferably configured or programmed to perform a control to switch between an engine driving state in which the propulsion generator is driven by the engine and an electric motor driving state in which the propulsion generator is driven by the electric motor. Accordingly, the controller functions as a controller for a hybrid system configured or programmed to perform a control to switch between the engine driving state and the electric motor driving state, and also functions as a controller configured or programmed to control the notifier. Consequently, as compared with a case in which the controller for the hybrid system is separated from the controller configured or programmed to control the notifier, a complex structure of the marine vessel including the marine power supply system is significantly reduced or prevented.

(13) A marine power supply system that switches between the engine driving state and the electric motor driving state preferably further includes a starter connected to the engine to start the engine, and a second battery provided separately from the first battery to supply power to the starter, and the controller is preferably configured or programmed to control the notifier to provide the notification indicating that the battery switch is in the off-state and to control the engine to start when power is supplied from the second battery to the starter and when the battery switch is in the off-state. Accordingly, the engine is started even when the notification indicating that the battery switch is in the off-state is provided. Consequently, even when the battery switch is in the off-state, the engine is started when the user desires to start the engine.

(14) A marine power supply system according to a preferred embodiment of the present invention preferably further includes a detector provided between the battery switch and the electric motor to detect that the battery switch is in the off-state. Accordingly, the user easily recognizes that the battery switch is in the off-state through the notifier based on the detection result detected by the detector.

(15) In such a case, a marine power supply system preferably further includes a power converter to convert DC power from the battery switch into AC power to supply the AC power to the electric motor, and a power conversion controller configured or programmed to control operation of the power converter and to also detect that the battery switch is in the off-state. Accordingly, it is not necessary to provide a dedicated detector to detect that the battery switch is in the off-state. Thus, an increase in the number of components of the marine power supply system is significantly reduced or prevented, and a complex structure of the marine power supply system is significantly reduced or prevented.

(16) A marine power supply system including the power conversion controller preferably further includes a third battery provided separately from the first battery to supply power to the power conversion controller. Accordingly, even when power is not supplied from the first battery to the power conversion controller, the power conversion controller detects that the battery switch is in the off-state using the power from the third battery.

(17) A marine power supply system including the detector preferably further includes a storage to store a detection result detected by the detector. Accordingly, the user or a maintenance worker is able to retroactively confirm the detection result stored in the storage.

(18) In a marine power supply system according to a preferred embodiment of the present invention, the first battery preferably includes a plurality of first batteries, the battery switch preferably includes a plurality of battery switches, and the notifier preferably provides a notification indicating that at least one of the plurality of battery switches is in the off-state when

the at least one of the plurality of battery switches is in the off-state. Accordingly, even when the plurality of battery switches are provided, the user is notified that at least one of the plurality of battery switches is in the off-state when at least one of the plurality of battery switches is in the off-state.

(19) A method for controlling a marine power supply system that supplies power to a marine propulsion unit including a propulsion generator that generates a propulsive force to propel a marine vessel, an engine that drives the propulsion generator, and an electric motor that drives the propulsion generator according to a preferred embodiment of the present invention includes determining that a battery switch to be manually switched between an on-state in which a first battery that supplies power to the electric motor is used and an off-state in which the first battery is not used is in the off-state, and providing a notification indicating that the battery switch is in the off-state based on the determining that the battery switch is in the off-state.

(20) In a method for controlling a marine power supply system according to a preferred embodiment of the present invention, similarly to the marine power supply system according to preferred embodiments of the present invention described above, even when power is supplied to the marine propulsion unit including the engine and the electric motor, traveling of the marine vessel without a user being aware that the battery switch is the off-state is significantly reduced or prevented.

(21) In a method for controlling the marine power supply system according to a preferred embodiment of the present invention, the providing of the notification preferably includes providing a display indicating that the battery switch is in the off-state when the battery switch is in the off-state. Accordingly, the user easily recognizes that the battery switch is in the off-state by visually recognizing the display.

(22) In such a case, the providing of the notification preferably includes displaying message information indicating that the battery switch is in the off-state when the battery switch is in the off-state. Accordingly, the user easily recognizes that the battery switch is in the off-state by reading the message information.

(23) In a method for controlling the marine power supply system according to a preferred embodiment of the present invention, the marine power supply system preferably further includes a starter connected to the engine to start the engine, and a second battery provided separately from the first battery to supply power to the starter, and the providing of the notification preferably includes providing the notification indicating that the battery switch is in the off-state and starting the engine when power is supplied from the second battery to the starter and when the battery switch is in the off-state. Accordingly, the engine is started even when the notification indicating that the battery switch is in the off-state is provided. Consequently, even when the battery switch is in the off-state, the engine is started when the user desires to start the engine.

(24) A method for controlling the marine power supply system according to a preferred embodiment of the present invention preferably further includes detecting that the battery switch is in the off-state, and the providing of the notification preferably includes providing the notification indicating that the battery switch is in the off-state based on detecting that the battery switch is in the off-state. Accordingly, it is detected that the battery switch is in the off-state. When it is detected that the battery switch is in the off-state, the user is notified that that the battery switch is in the off-state.

(25) In such a case, the detecting preferably includes detecting that the battery switch is in the off-state by using a power conversion controller configured or programmed to control operation of a power converter that converts DC power from the battery switch into AC power to supply the AC power to the electric motor. Accordingly, the power conversion controller detects that the battery switch is in the off-state without providing a detector separately from the power conversion controller.

(26) A method for controlling the marine power supply system including the detecting that the

battery switch is in the off-state by using the power conversion controller, the marine power supply system preferably further includes a third battery provided separately from the first battery to supply power to the power conversion controller, and the detecting preferably includes detecting that the battery switch is in the off-state by using the power conversion controller using the power from the third battery. Accordingly, even when power is not supplied from the first battery to the power conversion controller, the power conversion controller detects that the battery switch is in the off-state using the power from the third battery.

(27) A method for controlling the marine power supply system according to a preferred embodiment of the present invention preferably further includes storing a detected detection result after the detecting. Accordingly, the user or a maintenance worker is able to retroactively confirm the stored detection result.

(28) In a method for controlling the marine power supply system according to a preferred embodiment of the present invention, the first battery preferably includes a plurality of first batteries, the battery switch preferably includes a plurality of battery switches, and the providing of the notification preferably includes providing a notification indicating that at least one of the plurality of battery switches is in the off-state when the at least one of the plurality of battery switches is in the off-state. Accordingly, even when the plurality of battery switches are provided, the user is notified that at least one of the plurality of battery switches is in the off-state when at least one of the plurality of battery switches is in the off-state.

(29) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view schematically showing a marine vessel including a marine vessel control system according to a preferred embodiment of the present invention.

(2) FIG. 2 is a side view illustrating the structure of an outboard motor according to a preferred embodiment of the present invention.

(3) FIG. 3 is a block diagram showing the structure of a marine vessel according to a preferred embodiment of the present invention.

(4) FIG. 4 is a diagram showing an example of a message image according to a preferred embodiment of the present invention.

(5) FIG. 5 is another diagram showing an example of a message image according to a preferred embodiment of the present invention.

(6) FIG. 6 is still another diagram showing an example of a message image according to a preferred embodiment of the present invention.

(7) FIG. 7 is a diagram showing a display of detection results according to a preferred embodiment of the present invention.

(8) FIG. 8 is a flowchart showing a control process of a marine vessel control system according to a preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) Preferred embodiments of the present invention are hereinafter described with reference to the drawings.

(10) The structure of a marine vessel **100** according to preferred embodiments of the present invention is now described with reference to FIGS. 1 to 7. In the figures, arrow FWD represents the forward movement direction of the marine vessel **100**, and arrow BWD represents the reverse movement direction of the marine vessel **100**.

- (11) As shown in FIG. 1, the marine vessel **100** includes a hull **1**, outboard motors **2a** and **2b**, and a marine vessel control system **3**. That is, the marine vessel **100** is an outboard motor boat. Furthermore, the marine vessel control system **3** is a power supply system that supplies power to the outboard motors **2a** and **2b**. The outboard motors **2a** and **2b** are examples of a “marine propulsion unit”. The marine vessel control system **3** is an example of a “marine power supply system”.
- (12) The outboard motors **2a** and **2b** are attached to the hull **1**. For example, the outboard motors **2a** and **2b** are located side by side in a right-left direction. As shown in FIG. 2, the outboard motor **2a** includes an engine **21**, an electric motor **22**, a propeller **23**, an engine ECU (electronic control unit) **24**, and a starter motor **25**. The structure of the outboard motor **2b** is the same as that of the outboard motor **2a**. The propeller **23** is an example of a “propulsion generator”. The starter motor **25** is an example of a “starter”.
- (13) The engine **21** drives the propeller **23**. Specifically, the engine **21** is an internal combustion engine driven by combustion of gasoline, light oil, or the like. The engine **21** is connected to the propeller **23** via a drive shaft **21a**, etc. A driving force (rotational force) is transmitted from the engine **21** to the propeller **23** such that the propeller **23** rotates, and the marine vessel **100** (hull **1**) is propelled.
- (14) The electric motor **22** drives the propeller **23**. The electric motor **22** is a motor. The electric motor **22** provided in the outboard motor **2a** is driven using power from an electric motor battery **34a** described below. The electric motor **22** provided in the outboard motor **2b** is driven using power from an electric motor battery **34b** described below. Specifically, the electric motor **22** provided in the outboard motor **2a** is driven using power converted by an inverter device **5a** described below. The electric motor **22** provided in the outboard motor **2b** is driven using power converted by an inverter device **5b** described below. The electric motor **22** is connected to the propeller **23** via a shaft **22a**, a propeller shaft, etc. A driving force (rotational force) is transmitted from the electric motor **22** to the propeller **23** such that the propeller **23** rotates, and the marine vessel **100** (hull **1**) is propelled.
- (15) The propeller **23** is positioned in the water, and rotates to generate a propulsive force.
- (16) As shown in FIG. 3, the engine ECU **24** communicates with a hybrid controller **31** described below and the inverter device **5a** or **5b** provided in its own outboard motor **2a** or **2b**. For example, the engine ECU **24** transmits and receives information and command signals to and from the hybrid controller **31** and the inverter device **5a** or **5b** by control area network (CAN) communication. Furthermore, the engine ECUs **24** are connected to a main battery **35**, and operate using power from the main battery **35**.
- (17) The starter motors **25** start the engines **21**. Specifically, the starter motors **25** are connected to the main battery **35**, and are driven using power from the main battery **35**. The starter motors **25** rotate (crank) crankshafts of the engines **21** based on commands from the engine ECUs **24** (hybrid controller **31**). The engines **21** are cranked by the starter motors **25**, fuel is supplied into the engines **21** by fuel suppliers (not shown), and the fuel in the engines **21** is ignited by igniters (not shown) to start the engines **21**. Power is supplied from the main battery **35** to the fuel suppliers and the igniters.
- (18) The marine vessel control system **3** includes the hybrid controller **31**, a display **32**, a marine vessel maneuvering unit **33**, the electric motor batteries **34a** and **34b**, the main battery **35**, a house battery **36**, battery switches **4a** and **4b**, and inverter devices **5a** and **5b**. The hybrid controller **31** is an example of a “controller”. The display **32** is an example of a “notifier”. The electric motor batteries **34a** and **34b** are examples of a “first battery”. The main battery **35** is an example of a “second battery” or a “third battery”.
- (19) The hybrid controller **31**, the display **32**, the marine vessel maneuvering unit **33**, the electric motor batteries **34a** and **34b**, the main battery **35**, the house battery **36**, and the battery switches **4a** and **4b** are provided in the hull **1**, for example. The display **32** is positioned in the vicinity of the

marine vessel maneuvering unit **33**, for example. The display **32** is provided at a position visually recognized by a user who operates the marine vessel maneuvering unit **33**. On the display **32**, a gauge such as an engine speed is displayed. The inverter device **5a** is provided in the outboard motor **2a**. The inverter device **5b** is provided in the outboard motor **2b**.

(20) The battery switches **4a** and **4b** are manually switched between on-states in which the electric motor batteries **34a** and **34b** are used and off-states in which the electric motor batteries **34a** and **33b** are not used, respectively.

(21) Specifically, the battery switches **4a** and **4b** are connected in series between the electric motor battery **34a** and the inverter device **5a** and between the electric motor battery **34b** and the inverter device **5b**, respectively. The battery switch **4a** is rotated or moved by the user such that a state in which the electric motor battery **34a** and the inverter device **5a** are conducted (the electric motor battery **34a** is used) and a state in which the electric motor battery **34a** and the inverter device **5a** are shut off (the electric motor battery **34a** is not used) are switched. Furthermore, the battery switch **4b** is rotated or moved by the user such that a state in which the electric motor battery **34b** and the inverter device **5b** are conducted (the electric motor battery **34b** is used) and a state in which the electric motor battery **34b** and the inverter device **5b** are shut off (the electric motor battery **34b** is not used) are switched.

(22) The hybrid controller **31** is, for example, a control circuit that controls both driving of the engines **21** in the outboard motors **2a** and **2b** and driving of the electric motors **22** in the outboard motors **2a** and **2b**. For example, the hybrid controller **31** includes a processor such as a CPU to perform an arithmetic process and the like. Specifically, the hybrid controller **31** is configured or programmed to perform a control to switch between an engine driving state in which the propellers **23** are driven by the engines **21** and an electric motor driving state in which the propellers **23** are driven by the electric motors **22**. The hybrid controller **31** controls the outboard motors **2a** and **2b** in the electric motor driving state when the marine vessel **100** travels at a low speed, and controls the outboard motors **2a** and **2b** in the engine driving state when the marine vessel **100** travels at a high speed.

(23) As shown in FIGS. **4** to **6**, according to preferred embodiments of the present invention, the display **32** provides a notification indicating that at least one of the battery switch **4a** or **4b** is in an off-state when at least one of the battery switch **4a** or **4b** is in the off-state based on a command from the hybrid controller **31**. In other words, the hybrid controller **31** is configured or programmed to control the display **32** to provide a notification indicating that at least one of the battery switch **4a** or **4b** is in the off-state when at least one of the battery switch **4a** or **4b** is in the off-state.

(24) Specifically, as shown in FIG. **4**, the display **32** displays a message image **P1**, "first battery switch is in off-state", when the battery switch **4a** is in the off-state based on a command from the hybrid controller **31**. As shown in FIG. **5**, the display **32** displays a message image **P2**, "second battery switch is in off-state", when the battery switch **4b** is in the off-state based on a command from the hybrid controller **31**. As shown in FIG. **6**, the display **32** displays both the message images **P1** and **P2** when both the battery switches **4a** and **4b** are in the off-states based on a command from the hybrid controller **31**. The message images **P1** and **P2** are examples of "message information".

(25) According to preferred embodiments of the present invention, the hybrid controller **31** is configured or programmed to control the display **32** to provide a notification indicating that at least one of the battery switch **4a** or **4b** is in the off-state and to control the engines **21** to start when power is supplied from the main battery **35** to the starter motors **25** and when at least one of the battery switch **4a** or **4b** is in the off-state. That is, the marine vessel **100** (both the outboard motors **2a** and **2b**) is able to start the engines **21** even when at least one of the battery switch **4a** or **4b** is in the off-state.

(26) As shown in FIG. **3**, the marine vessel maneuvering unit **33** includes a remote control, for example. The marine vessel maneuvering unit **33** receives input operations by the user and transmits operation signals based on the received input operations to the hybrid controller **31**. The

hybrid controller **31** controls the operation of the outboard motors **2a** and **2b** based on the operation signals.

(27) The electric motor battery **34a**, the electric motor battery **34b**, the main battery **35**, and the house battery **36** are separate from each other. The electric motor battery **34a** supplies power to the electric motor **22** via the inverter device **5a**. The electric motor battery **34b** supplies power to the electric motor **22** via the inverter device **5b**. The main battery **35** supplies power to each of the starter motors **25** that start the engines **21** of the outboard motors **2a** and **2b**. The house battery **36** supplies power to at least some devices (devices in house) other than the electric motors **22** and the starter motors **25** among the devices in the marine vessel **100**. The electric motor batteries **34a** and **34b**, the main battery **35**, and the house battery **36** are, for example, lead storage batteries or lithium-ion batteries.

(28) The inverter devices **5a** and **5b** convert DC power from the electric motor batteries **34a** and **34b** into AC power based on commands from at least one of the hybrid controller **31** or the engine ECUs **24** to supply the AC power to the electric motors **22** of the outboard motors **2a** and **2b** in which the inverter devices **5a** and **5b** are located, respectively. The inverter device **5a** is positioned between the battery switch **4a** and the electric motor **22** of the outboard motor **2a**. The inverter device **5b** is positioned between the battery switch **4b** and the electric motor **22** of the outboard motor **2b**.

(29) The inverter device **5a** includes a power converter **51**, an inverter controller **52**, and a storage **53**. The structure of the inverter device **5b** is the same as that of the inverter device **5a**. The inverter controller **52** is an example of a “power conversion controller”.

(30) Each of the power converters **51** includes a switching circuit. Each of the power converters **51** converts DC power from the battery switch **4a** or **4b** corresponding to its own outboard motor **2a** or **2b** into AC power to supply the AC power to the corresponding electric motor **22**.

(31) Each of the inverter controllers **52** includes a microcomputer. The inverter controllers **52** are configured or programmed to control the operation of the power converters **51**. The inverter controller **52** provided in the outboard motor **2a** is configured or programmed to detect that the battery switch **4a** is in the off-state. Specifically, the inverter controller **52** detects that the battery switch **4a** is in an on-state based on the electric motor battery **34a** supplying power to the power converter **51** (a voltage being applied, for example), and detects that the battery switch **4a** is in the off-state based on the electric motor battery **34a** supplying no power to the power converter **51** (no voltage being applied, for example).

(32) Each of the storages **53** includes a non-volatile memory. Each of the storages **53** stores the date and time and information (detection result) indicating whether the battery switch **4a** or **4b** is in the off-state or the battery switch **4a** or **4b** is in the on-state in association with each other based on a command from the inverter controller **52**. As shown in FIG. 7, the hybrid controller **31** reads information in which the date and time and the detection result are associated with each other by accessing each of the storages **53**, and controls the display **32** to display an image P3, which is the information in which the date and time and the detection result are associated with each other.

(33) A method for controlling the marine vessel **100** according to preferred embodiments of the present invention is now described with reference to FIG. 8. FIG. 8 shows a flowchart of a control process in the marine vessel **100**. The following control process is executed by the hybrid controller **31** and the inverter controllers **52**.

(34) In step S1, the inverter controller **52** of the outboard motor **2a** acquires the detection result of whether or not the battery switch **4a** is in the off-state, and the inverter controller **52** of the outboard motor **2b** acquires the detection result of whether or not the battery switch **4b** is in the off-state. Then, the process advances to step S2.

(35) In step S2, the hybrid controller **31** determines whether or not at least one of the battery switch **4a** or **4b** is in the off-state. When at least one of the battery switch **4a** or **4b** is in the off-state, the process advances to step S3, and when each of the battery switches **4a** and **4b** is not in the off-state

(is in the on-state), the process returns to step S1.

(36) In step S3, at least one of the message image P1 indicating that the battery switch 4a is in the off-state or the message image P2 indicating that the battery switch 4b is in the off-state is displayed on the display 32. Then, the process returns to step S1.

(37) When power is supplied from the main battery 35 to the starter motors 25, the engines 21 are started even when at least one of the message image P1 or P2 is displayed.

(38) In the structure according to the various preferred embodiments described above, the following advantageous effects are achieved.

(39) According to a preferred embodiment of the present invention, the marine vessel control system 3 includes the display 32 to provide a notification indicating that at least one of the battery switch 4a or 4b is in the off-state when at least one of the battery switch 4a or 4b is in the off-state. Accordingly, the display 32 notifies the user that at least one of the battery switch 4a or 4b is in the off-state. Consequently, even when power is supplied to the outboard motors 2a and 2b including the engines 21 and the electric motors 22, traveling of the marine vessel 100 without the user being aware that the battery switch 4a or 4b is in the off-state is significantly reduced or prevented.

(40) According to a preferred embodiment of the present invention, the marine vessel control system 3 includes the hybrid controller 31 configured or programmed to control the display 32 to provide a notification indicating that at least one of the battery switch 4a or 4b is in the off-state when at least one of the battery switch 4a or 4b is in the off-state. Accordingly, the display 32 is easily controlled using the hybrid controller 31.

(41) According to a preferred embodiment of the present invention, the hybrid controller 31 is configured or programmed to control the display 32 to provide a display indicating that at least one of the battery switch 4a or 4b is in the off-state when at least one of the battery switch 4a or 4b is in the off-state. Accordingly, the user easily recognizes that at least one of the battery switch 4a or 4b is in the off-state by visually recognizing the display 32.

(42) According to a preferred embodiment of the present invention, the hybrid controller 31 is configured or programmed to control the display 32 to display at least one of the message image P1 or P2 indicating that at least one of the battery switch 4a or 4b is in the off-state when at least one of the battery switch 4a or 4b is in the off-state. Accordingly, the user easily recognizes that at least one of the battery switch 4a or 4b is in the off-state by visually recognizing at least one of the message image P1 or P2 on the display 32 and reading the message content.

(43) According to a preferred embodiment of the present invention, the hybrid controller 31 is configured or programmed to perform a control to switch between the engine driving state in which the propellers 23 are driven by the engines 21 and the electric motor driving state in which the propellers 23 are driven by the electric motors 22. Accordingly, the hybrid controller 31 defines and functions as a controller for a hybrid system configured or programmed to perform a control to switch between the engine driving state and the electric motor driving state, and also defines and functions as a controller configured or programmed to control the display 32. Consequently, as compared with a case in which the controller for the hybrid system is separate from the controller configured or programmed to control the display 32, a complex structure of the marine vessel 100 including the marine vessel control system 3 is significantly reduced or prevented.

(44) According to a preferred embodiment of the present invention, the marine vessel control system 3 includes the starter motors 25 connected to the engines 21 to start the engines 21, and the main battery 35 provided separately from the electric motor batteries 34a and 34b to supply power to the starter motors 25. Furthermore, the hybrid controller 31 is configured or programmed to control the display 32 to provide a notification indicating that at least one of the battery switch 4a or 4b is in the off-state and to control the engines 21 to start when power is supplied from the main battery 35 to the starter motors 25 and when at least one of the battery switch 4a or 4b is in the off-state. Accordingly, the engines 21 are started even when the notification indicating that at least one of the battery switch 4a or 4b is in the off-state is provided. Consequently, even when at least one

of the battery switch **4a** or **4b** is in the off-state, the engines **21** are started when the user desires to start the engines **21**.

(45) According to a preferred embodiment of the present invention, the marine vessel control system **3** includes the inverter devices **5a** and **5b** (inverter controllers **52**) provided between the battery switches **4a** and **4b** and the electric motors **22** to detect that at least one of the battery switch **4a** or **4b** is in the off-state. Accordingly, the user easily recognizes that at least one of the battery switch **4a** or **4b** is in the off-state through the display **32** based on the detection results detected by the inverter controllers **52**.

(46) According to a preferred embodiment of the present invention, the marine vessel control system **3** includes the power converters **51** to convert the DC power from the battery switches **4a** and **4b** into AC power to supply the AC power to the electric motors **22**, and the inverter controllers **52** configured or programmed to control the operation of the power converters **51**. Accordingly, it is not necessary to provide a dedicated controller to detect that at least one of the battery switch **4a** or **4b** is in the off-state. Thus, an increase in the number of components of the marine vessel control system **3** is significantly reduced or prevented, and a complex structure of the marine vessel control system **3** is significantly reduced or prevented.

(47) According to a preferred embodiment of the present invention, the marine vessel control system **3** includes the main battery **35** provided separately from the electric motor batteries **34a** and **34b** to supply power to the inverter controllers **52**. Accordingly, even when power is not supplied from the electric motor batteries **34a** and **34b** to the inverter controllers **52**, the inverter controllers **52** detect that at least one of the battery switch **4a** or **4b** is in the off-state using the power from the main battery **35**.

(48) According to a preferred embodiment of the present invention, the marine vessel control system **3** includes the storages **53** to store the detection results detected by the inverter controllers **52**. Accordingly, the user or a maintenance worker is able to retroactively confirm the detection results stored in the storages **53**.

(49) According to a preferred embodiment of the present invention, the marine vessel control system **3** includes the electric motor batteries **34a** and **34b**, and the battery switches **4a** and **4b**. Furthermore, the display **32** provides a notification indicating that at least one of the battery switch **4a** or **4b** is in the off-state when at least one of a plurality of battery switches **4a** and **4b** is in the off-state. Accordingly, even when the battery switches **4a** and **4b** are provided, the user is notified that at least one of the battery switch **4a** or **4b** is in the off-state when at least one of the battery switch **4a** or **4b** is in the off-state.

(50) In the control method according to the various preferred embodiments described above, the following advantageous effects are achieved.

(51) According to a preferred embodiment of the present invention, even when power is supplied to the outboard motors **2a** and **2b** including the engines **21** and the electric motors **22**, traveling of the marine vessel **100** without the user being aware that the battery switch **4a** or **4b** is the off-state is significantly reduced or prevented, similarly to the structure of the marine vessel control system **3**.

(52) The preferred embodiments of the present invention described above are illustrative in all points and not restrictive. The extent of the present invention is not defined by the above description of the preferred embodiments but by the scope of the claims, and all modifications within the meaning and range equivalent to the scope of the claims are further included.

(53) For example, while the marine vessel is preferably an outboard motor boat in the preferred embodiments described above, the present invention is not restricted to this. That is, the marine vessel may alternatively be a marine vessel other than an outboard motor boat. For example, the marine vessel may be a marine vessel including an inboard motor(s), an inboard-outboard motor(s), or a jet propulsion device(s).

(54) While the main battery, the electric motor batteries, and the house battery are preferably provided in the hull in the preferred embodiments described above, the present invention is not

restricted to this. That is, at least one of the main battery, the electric motor batteries, or the house battery may alternatively be provided in the outboard motors.

(55) While power is preferably supplied from the electric motor batteries only to the electric motors via the inverter devices in the preferred embodiments described above, the present invention is not restricted to this. That is, power may alternatively be supplied from the electric motor batteries to other devices in addition to the electric motors.

(56) While two battery switches are preferably provided in the preferred embodiments described above, the present invention is not restricted to this. That is, one common battery switch may alternatively be provided for a plurality of electric motor batteries, or three or more battery switches may alternatively be provided.

(57) While the display preferably provides a notification indicating that the battery switch is in the off-state in the preferred embodiments described above, the present invention is not restricted to this. For example, a sound output may alternatively be provided in the marine vessel, and the sound output may alternatively provide a notification indicating that the battery switch is in the off-state.

(58) While the display preferably displays the message image indicating that the battery switch is in the off-state in the preferred embodiments described above, the present invention is not restricted to this. For example, the display may alternatively display a warning light indicating that the battery switch is in the off-state without using a character image.

(59) While the hybrid controller preferably controls the display as a notifier in the preferred embodiments described above, the present invention is not restricted to this. For example, the marine vessel control system may alternatively include a controller dedicated for the display as a notifier separately from the hybrid controller.

(60) While each of the inverter controllers preferably detects that the battery switch is in the off-state in the preferred embodiments described above, the present invention is not restricted to this. For example, a detector may alternatively be provided to detect that the battery switch is in the off-state by detecting a voltage or current in a path between the battery switch and the inverter device, or a detector may alternatively be provided to detect that the battery switch is in the off-state by detecting the state (such as the rotation angle or position) of the battery switch.

(61) While power is preferably supplied from the main battery to the inverter controllers in the preferred embodiments described above, the present invention is not restricted to this. For example, power may alternatively be supplied from the house battery to the inverter controllers.

(62) While the storages are preferably provided in the inverter devices in the preferred embodiments described above, the present invention is not restricted to this. For example, the storages may alternatively be provided in the hull.

(63) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A marine power supply system that supplies power to a marine propulsion unit including a propulsion generator that generates a propulsive force to propel a marine vessel, an engine that drives the propulsion generator, and an electric motor that drives the propulsion generator, the marine power supply system comprising: a first battery to supply power to the electric motor; a battery switch to be manually switched between an on-state in which the first battery is used and an off-state in which the first battery is not used; and a notifier to provide a notification indicating that the battery switch is in the off-state when the battery switch is in the off-state without providing a notification to the user indicating that the battery switch is in the on-state when the battery switch is in the on-state.

2. The marine power supply system according to claim 1, further comprising: a controller configured or programmed to control the notifier to provide the notification indicating that the battery switch is in the off-state when the battery switch is in the off-state.
3. The marine power supply system according to claim 2, wherein the notifier includes a display; and the controller is configured or programmed to control the display to provide a display indicating that the battery switch is in the off-state when the battery switch is in the off-state.
4. The marine power supply system according to claim 3, wherein the controller is configured or programmed to control the display to display message information indicating that the battery switch is in the off-state when the battery switch is in the off-state.
5. The marine power supply system according to claim 2, wherein the controller is configured or programmed to perform a control to switch between an engine driving state in which the propulsion generator is driven by the engine and an electric motor driving state in which the propulsion generator is driven by the electric motor.
6. The marine power supply system according to claim 5, further comprising: a starter connected to the engine to start the engine; and a second battery provided separately from the first battery to supply power to the starter; wherein the controller is configured or programmed to control the notifier to provide the notification indicating that the battery switch is in the off-state and to control the engine to start when power is supplied from the second battery to the starter and when the battery switch is in the off-state.
7. The marine power supply system according to claim 1, further comprising: a detector provided between the battery switch and the electric motor to detect that the battery switch is in the off-state.
8. The marine power supply system according to claim 7, further comprising: a power converter to convert DC power from the battery switch into AC power to supply the AC power to the electric motor; and a power conversion controller configured or programmed to control operation of the power converter and to also define and function as the detector.
9. The marine power supply system according to claim 8, further comprising: a third battery provided separately from the first battery to supply power to the power conversion controller.
10. The marine power supply system according to claim 7, further comprising: a storage to store a detection result detected by the detector.
11. The marine power supply system according to claim 1, wherein the first battery includes a plurality of first batteries; the battery switch includes a plurality of battery switches; and the notifier provides a notification indicating that at least one of the plurality of battery switches is in the off-state when the at least one of the plurality of battery switches is in the off-state.
12. A method for controlling a marine power supply system that supplies power to a marine propulsion unit including a propulsion generator that generates a propulsive force to propel a marine vessel, an engine that drives the propulsion generator, and an electric motor that drives the propulsion generator, the marine power supply system including a starter connected to the engine to start the engine and a second battery provided separately from the first battery to supply power to the starter, the method comprising: determining that a battery switch to be manually switched between an on-state in which a first battery that supplies power to the electric motor is used and an off-state in which the first battery is not used is in the off-state by a controller; and providing a notification indicating that the battery switch is in the off-state based on the determining that the battery switch is in the off-state without providing a notification to the user indicating that the battery switch is in the on-state when the battery switch is in the on-state; wherein the providing the notification includes providing the notification indicating that the battery switch is in the off-state, and starting the engine when power is supplied from the second battery to the starter and the battery switch is in the off-state.
13. The method according to claim 12, wherein the providing of the notification includes providing a display indicating that the battery switch is in the off-state when the battery switch is in the off-state.

14. The method according to claim 13, wherein the providing of the notification includes displaying message information indicating that the battery switch is in the off-state when the battery switch is in the off-state.

15. The method according to claim 12, further comprising: detecting that the battery switch is in the off-state; wherein the providing of the notification includes providing the notification indicating that the battery switch is in the off-state based on detecting that the battery switch is in the off-state.

16. The method according to claim 15, wherein the detecting includes detecting that the battery switch is in the off-state with a power conversion controller configured or programmed to control operation of a power converter that converts DC power from the battery switch into AC power to supply the AC power to the electric motor.

17. The method according to claim 16, wherein the marine power supply system further includes a third battery provided separately from the first battery to supply power to the power conversion controller; and the detecting includes detecting that the battery switch is in the off-state by using the power conversion controller using the power from the third battery.

18. The method according to claim 15, further comprising: storing a detected detection result after the detecting.

19. The method according to claim 12, wherein the first battery includes a plurality of first batteries; the battery switch includes a plurality of battery switches; and the providing of the notification includes providing a notification indicating that at least one of the plurality of battery switches is in the off-state when the at least one of the plurality of battery switches is in the off-state.

20. A method for controlling a marine power supply system that supplies power to a marine propulsion unit including a propulsion generator that generates a propulsive force to propel a marine vessel, an engine that drives the propulsion generator, and an electric motor that drives the propulsion generator, the method comprising: determining that a battery switch to be manually switched between an on-state in which a first battery that supplies power to the electric motor is used and an off-state in which the first battery is not used is in the off-state by a controller; providing a notification indicating that the battery switch is in the off-state based on the determining that the battery switch is in the off-state without providing a notification to the user indicating that the battery switch is in the on-state when the battery switch is in the on-state; and detecting that the battery switch is in the off-state with a power conversion controller configured or programmed to control operation of a power converter that converts DC power from the battery switch into AC power to supply the AC power to the electric motor.
